Strategic Behavior

Midterm Exam: Solutions and Grading Guide • Spring 2024

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General grading policy. Each question lists the points for each step. When a student sets up the game correctly but makes an algebra slip, deduct only at the slip and grade the rest for consistency with their numbers. Economic interpretations must be consistent with the student's own computations. Monetary quantities in Question 1 are in thousands of dollars.

Question 1: The Cement Duopoly (14 points)

(a) Rocklin's profit is $\pi_1 = (120 - q_1 - q_2 - 12)\,q_1 = (108 - q_1 - q_2)\,q_1$. The first-order condition $108 - 2q_1 - q_2 = 0$ and the second-order condition $-2 < 0$ give

$$BR_1(q_2) = \max\left\{\tfrac{108 - q_2}{2},\, 0\right\}.$$

Similarly $\pi_2 = (96 - q_1 - q_2)\,q_2$ gives

$$BR_2(q_1) = \max\left\{\tfrac{96 - q_1}{2},\, 0\right\}.$$

> **Grading.** 2 pts each best response: 1 pt for the profit function and FOC, 1 pt for the solved form. Omitting the $\max\{\cdot, 0\}$ is not penalized here.

(b) Solving $2q_1 + q_2 = 108$ and $q_1 + 2q_2 = 96$ gives $q_1 = 40$, $q_2 = 28$. Then $Q = 68$ and $P = 52$.

$$\pi_1 = (52 - 12) \cdot 40 = 1600, \qquad \pi_2 = (52 - 24) \cdot 28 = 784.$$

> **Grading.** 2 pts: quantities. 1 pt: price. 2 pts: profits (1 each). A student who solves the symmetric-cost model by mistake (e.g. $c_1 = c_2 = 12$) earns at most 2/5.

(c) With a general c_2, $BR_2(q_1) = (120 - c_2 - q_1)/2$. Substituting BR_1 yields

$$q_2 = \frac{132 - 2c_2}{3},$$

which is positive iff $c_2 < 66$. If $c_2 \geq 66$, Tidewater produces nothing and Rocklin is a monopolist: $q_1 = \frac{108}{2} = 54$ and $P = 66 \leq c_2$, so Tidewater cannot sell at a price above its cost (its best response to $q_1 = 54$ is $\max\{(66 - c_2)/2, 0\} = 0$).

> **Grading.** 2 pts: threshold $c_2 < 66$ from the interior solution. 1 pt: monopoly outcome $q_1 = 54$, $P = 66$ for $c_2 \geq 66$.

(d) Consumer surplus is the triangle between demand and the price: $\frac{1}{2}(120-52)\cdot 68 = \boxed{2312}$.

> **Grading.** 1 pt: formula. 1 pt: value.

Question 2: Where Should the Candidates Stand? (12 points)

(a) With $x_1 < x_2$, every voter whose ideal point is below the midpoint $(x_1+x_2)/2$ is closer to candidate 1, so candidate 1's vote share is

$$\boxed{F\left(\tfrac{x_1+x_2}{2}\right) = \left(\tfrac{x_1+x_2}{2}\right)^2}.$$

> **Grading.** 2 pts: midpoint argument. 1 pt: correct use of $F(x) = x^2$ (not the uniform CDF).

(b) $F(m) = m^2 = \frac{1}{2}$, so $m = \frac{1}{\sqrt{2}} \approx 0.707$.

> **Grading.** 2 pts. Giving $m = \frac{1}{2}$ (uniform median) earns 0; carry the student's m forward in (c) and (d).

(c) At (m,m) the vote is tied and each candidate gets $\frac{1}{2}$. If candidate 1 moves to $x < m$, her share is $F\left(\frac{x+m}{2}\right) < F(m) = \frac{1}{2}$ because $\frac{x+m}{2} < m$ and F is strictly increasing; she loses and gets 0. If she moves to $x > m$, her share is $1 - F\left(\frac{x+m}{2}\right) < 1 - F(m) = \frac{1}{2}$; again she loses. No deviation is profitable; by symmetry the same holds for candidate 2.

> **Grading.** 1 pt: payoff $\frac{1}{2}$ at (m,m). 2 pts: both directions of deviation.

(d) Take any profile other than (m,m).

- *Same platform $x \neq m$.* Each gets $\frac{1}{2}$. If $x < m$, candidate 1 moves to m and wins the voters to the right of $\frac{x+m}{2} < m$, a share $1 - F\left(\frac{x+m}{2}\right) > \frac{1}{2}$; she wins and gets $1 > \frac{1}{2}$. The case $x > m$ is symmetric.
- *Different platforms, $x_1 < x_2$, with a loser.* The loser gets 0 and can move to the winner's platform to force a tie, earning $\frac{1}{2} > 0$.
- *Different platforms, $x_1 < x_2$, with a tie.* Then $F\left(\frac{x_1+x_2}{2}\right) = \frac{1}{2}$, so the midpoint equals m and $x_1 < m < x_2$. Candidate 1 moves to m; her share becomes $F\left(\frac{m+x_2}{2}\right) > F(m) = \frac{1}{2}$, so she wins and gets $1 > \frac{1}{2}$.

In every case some candidate has a profitable deviation, so (m,m) is the unique Nash equilibrium.

> **Grading.** 1 pt: same-platform case. 1 pt: loser case. 2 pts: tie with different platforms (the case most students miss). A purely verbal "they converge to the median" argument earns at most 1/4.

Question 3: Who Calls the Utility? (14 points)

(a) In any pure equilibrium exactly one household phones. If nobody phones, any household gains $v - c = 9 > 0$ by phoning. If two or more phone, any caller saves c by not phoning, since the line is still repaired. If exactly one phones, the caller gets 9 versus 0 from hanging up, and each other household gets 12 versus 9 from also phoning. There are n pure equilibria, one for each choice of the caller.

> **Grading.** 1 pt: rules out zero callers. 1 pt: rules out two or more. 1 pt: verifies exactly one caller (and counts n).

(b) Let each household not phone with probability ρ. A household that phones gets $v - c$ for sure. A household that does not phone gets v unless all $n-1$ others also do not phone, so its expected payoff is $v\left(1-\rho^{n-1}\right)$. The symmetric pure profiles (all phone, nobody phones) are not equilibria by (a), so in a symmetric equilibrium $0 < \rho < 1$ and each household is indifferent:

$$v - c = v\left(1-\rho^{n-1}\right) \iff \rho^{n-1} = \frac{c}{v} \iff \rho = \left(\frac{c}{v}\right)^{1/(n-1)} = \left(\frac{1}{4}\right)^{1/(n-1)}.$$

> **Grading.** 2 pts: payoff from not phoning. 2 pts: indifference condition. 1 pt: solution for ρ.
> Common error: writing the payoff from not phoning as $v(1-\rho^n)$ (including one's own choice): award 3/5.

(c) $n = 2$: $\rho = \frac{1}{4}$ and $\Pr(\text{nobody phones}) = \rho^2 = \frac{1}{16} = 0.0625$.
$n = 3$: $\rho = \left(\frac{1}{4}\right)^{1/2} = \frac{1}{2}$ and $\Pr(\text{nobody phones}) = \rho^3 = \frac{1}{8} = 0.125$.

> **Grading.** 1 pt: both values of ρ. 2 pts: both probabilities that nobody phones.

(d) $\Pr(\text{nobody phones}) = \rho^n = \left(\frac{c}{v}\right)^{n/(n-1)}$. Since $n/(n-1)$ decreases to 1 and $c/v < 1$, this probability *increases* in n and converges to $\frac{c}{v} = \frac{1}{4}$. More witnesses make each one so much less willing to act that the line is *less* likely to be reported (a "bystander effect"). Each household's equilibrium payoff stays $v - c = 9$ for every n.

> **Grading.** 1 pt: expression ρ^n. 1 pt: limit $\frac{1}{4}$. 1 pt: interpretation (direction must be "increases").

Question 4: Ferry Pricing with Limited Seats (12 points)

(a) At $p = 60$, $D(60) = 40$; each operator sells $\min\{20, 20\} = 20$, so both ferries are full. Revenue $= 60 \cdot 20 =$ 1200 each.

> **Grading.** 2 pts: sales of 20 each (full). 1 pt: revenue.

(b) Fix $p_2 = 60$ and consider operator 1.

- *Undercutting*, $p < 60$: operator 1 is cheaper and sells $\min\{20, 100 - p\} = 20$ seats, so revenue is $20p < 1200$. Lowering price cannot raise sales beyond capacity.
- *Raising the price*, $60 < p \leq 80$: operator 1 serves the residual demand $D(p) - 20 = 80 - p < 20$, so revenue is $R(p) = p(80 - p)$. This is a downward parabola with maximum at $p = 40$, hence strictly decreasing on $[60, 80]$; so $R(p) < R(60) = 60 \cdot 20 = 1200$. For $p > 80$ operator 1 sells nothing.

No deviation is profitable, and by symmetry $(60, 60)$ is a Nash equilibrium.

> **Grading.** 2 pts: downward deviations. 3 pts: upward deviations using the residual demand $80 - p$ and the fact that its revenue peaks at $40 < 60$. Checking only one or two specific prices (e.g. $p = 61$) earns at most 1 of these 3 pts.

(c) At (p^*, p^*) each ferry is full, so undercutting only lowers revenue, exactly as in (b). An operator that raises its price to $p \geq p^*$ earns $R(p) = p\,(100 - k - p)$, which is maximized at $p_r = \frac{100-k}{2}$; at $p = p^*$ the residual demand equals $100 - k - p^* = k$, so R is continuous there.

- If $p_r \leq p^*$, R is decreasing on $[p^*, 100 - k]$ and no upward deviation pays.
- If $p_r > p^*$, moving to p_r strictly raises revenue.

$p_r \leq p^* \iff \frac{100-k}{2} \leq 100 - 2k \iff 3k \leq 100$. Hence (p^*, p^*) is a Nash equilibrium iff $0 < k \leq \frac{100}{3}$. (For example, with $k = 40$: $p^* = 20$ and revenue 800, but deviating to $p_r = 30$ earns $30 \cdot 30 = 900$.)

> **Grading.** 1 pt: undercutting unprofitable. 2 pts: residual-demand revenue and its maximizer p_r. 1 pt: condition $k \leq \frac{100}{3}$.

Question 5: Penalty Kicks (8 points)

(a) Let $p = \Pr(\text{kicker aims L})$ and $q = \Pr(\text{goalie dives L})$. The kicker's mix makes the goalie indifferent:

$$0.55p + 0.90(1-p) = 0.95p + 0.60(1-p) \iff 0.90 - 0.35p = 0.60 + 0.35p \iff p = \tfrac{3}{7}.$$

The goalie's mix makes the kicker indifferent:

$$0.55q + 0.95(1-q) = 0.90q + 0.60(1-q) \iff 0.95 - 0.40q = 0.60 + 0.30q \iff q = \tfrac{1}{2}.$$

Goal probability: $0.95 - 0.40 \cdot \frac{1}{2} = 0.75$.

Grading. 1.5 pts: $p = \frac{3}{7}$. 1.5 pts: $q = \frac{1}{2}$. 1 pt: goal probability. Swapping p and q (solving each player's mix from her own payoffs): at most 2/4.

(b) Goalie indifferent: $0.90 - 0.35p = 0.95p + 0.70(1-p) = 0.70 + 0.25p \iff p = \frac{1}{3}$. Kicker indifferent: $0.95 - 0.40q = 0.90q + 0.70(1-q) = 0.70 + 0.20q \iff q = \frac{5}{12}$. Goal probability $0.95 - 0.40 \cdot \frac{5}{12} = \frac{47}{60} \approx 0.783$.

The kicker now aims **Right more often** ($\frac{2}{3}$ instead of $\frac{4}{7}$), and the goalie dives **Right more often** ($\frac{7}{12}$ instead of $\frac{1}{2}$). In a mixed equilibrium each player's mixture is pinned down by the *opponent's* indifference. Aiming Right has become more attractive to the kicker, so the kicker stays indifferent only if the goalie guards Right more often. Diving Right has become costlier for the goalie, so the goalie stays indifferent only if diving Left becomes costlier too, which happens when the kicker aims Right (where a Left dive concedes 0.90) more often.

Grading. 1 pt: new p. 1 pt: new q. 1 pt: directions and new goal probability. 1 pt: explanation based on opponents' indifference.